CSA5108 Cryptography and Network security

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Exp 13 : Known-Plaintext Attack on the Hill Cipher
Code:

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#include <stdio.h>

int mod(int x) {
   return (x % 26 + 26) % 26;
}

int inverse(int x) {
   for (int i=1;i<26;i++)
      if (mod(x*i)==1) return i;
   return -1;
}

int main(void) {
   char p[5], c[5];
   int P[2][2], C[2][2], inv, det;
   int K[2][2];

   printf("Enter 4 plaintext letters: ");
   scanf("%4s",p);
   printf("Enter corresponding 4 ciphertext letters: ");
   scanf("%4s",c);

   for(int i=0;i<4;i++) {
      if(p[i]<'A'||p[i]>'Z'||c[i]<'A'||c[i]>'Z') {
         printf("Use uppercase letters only.\n");
         return 1;
      }
   }

   P[0][0]=p[0]-'A'; P[1][0]=p[1]-'A';
```

```
    P[0][1]=p[2]-'A'; P[1][1]=p[3]-'A';
    C[0][0]=c[0]-'A'; C[1][0]=c[1]-'A';
    C[0][1]=c[2]-'A'; C[1][1]=c[3]-'A';

    det=mod(P[0][0]*P[1][1]-P[0][1]*P[1][0]);
    inv=inverse(det);

    if(inv==-1) {
        printf("Plaintext matrix is not invertible modulo 26.\n");
        return 1;
    }
    int Q[2][2]={
        {mod(P[1][1]*inv),mod(-P[0][1]*inv)},
        {mod(-P[1][0]*inv),mod(P[0][0]*inv)}
    };
    for(int i=0;i<2;i++)
        for(int j=0;j<2;j++)
            K[i][j]=mod(C[i][0]*Q[0][j]+C[i][1]*Q[1][j]);

    printf("Recovered key matrix:\n");
    printf("%d %d\n%d %d\n",K[0][0],K[0][1],K[1][0],K[1][1]);

    return 0;
}
```

